



Mansoura University
Faculty of Computers and Information
2025/2026



MODELLING & SIMULATION

SW,BI,AI

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COURSE SYLLABUS

- **Chapter 1: Introduction.**
- **Chapter 2: Probability as Using in Simulation.**
- **Chapter 3: Queueing Simulation.**
- **Chapter 4: Inventory Simulation.**
- **Chapter 5: Random-Number Generation.**
- **Chapter 6: Input Modeling.**
- **Chapter 7: Output Analysis.**

REFERENCES

Simulation Modeling and Analysis

Averill M. Law
Fifth Edition

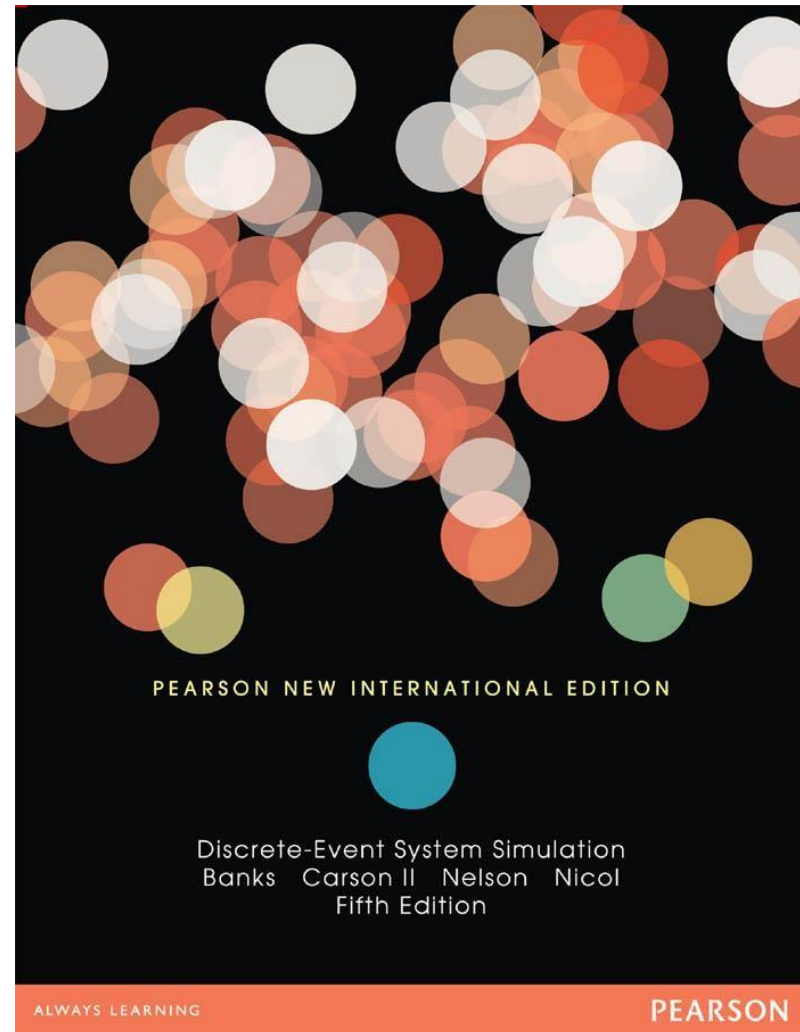
SIMULATION MODELING AND ANALYSIS

5^e

Averill M. Law

REFERENCES

- **Discrete-Event
System Simulation**
- Banks Carson II
Nelson Nicol

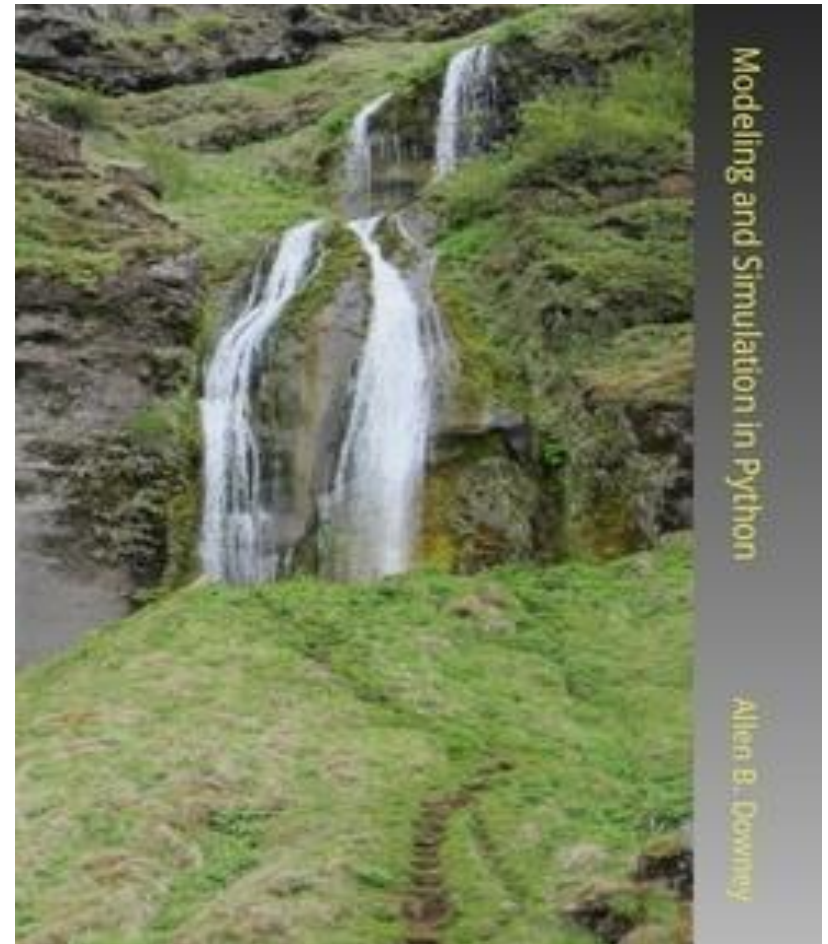


REFERENCES

Modeling and Simulation in Python

Allen B. Downey

Version 1.0.2



Simulation Software

- AnyLogic is a professional simulation tool used in industry and academia



**You can find documents and
tutorials at this website**



<https://www.anylogic.com/>

CHAPTER I: INTRODUCTION

- **General Introduction.**
 - ✓ **Important Definitions.**
 - ✓ **When Simulation is the Appropriate Tool.**
 - ✓ **When Simulation is Not Appropriate.**
 - ✓ **Advantages and Disadvantages of Simulation.**
- **Areas of Application.**
- **Systems and System Environment.**
- **Components of a System.**
- **Classification of Systems.**
- **Steps in a Simulation Study.**

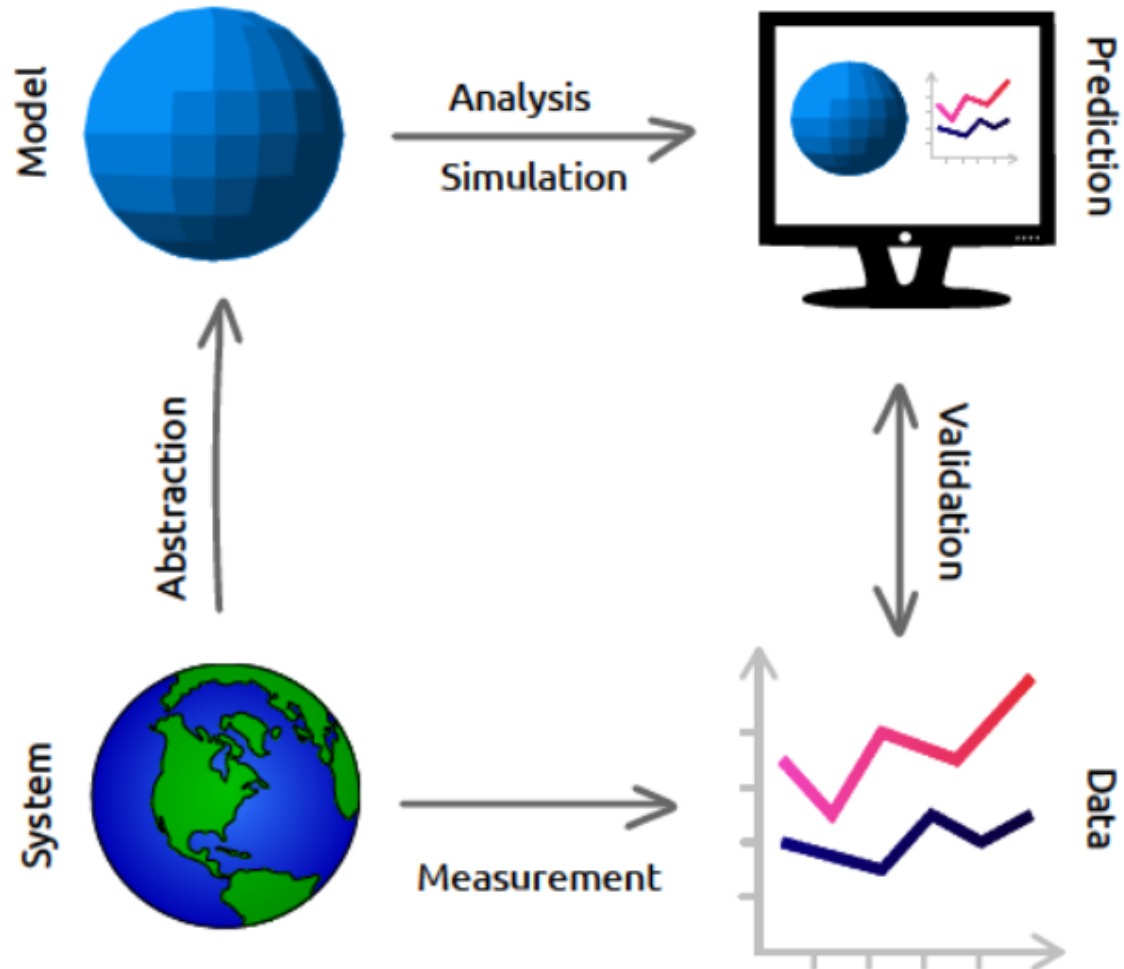
CHAPTER I: INTRODUCTION

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 - ✓ **Important Definitions.**
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 - ✓ **When Simulation is Not Appropriate.**
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- **Steps in a Simulation Study.**

COURSE OBJECTIVES

- Using computers to imitate, or *simulate*, the operations of various kinds of real-world facilities or processes.

MODELING AND SIMULATION OF PHYSICAL SYSTEMS



GENERAL INTRODUCTION

- **System:** Something in the real world we're interested in studying. We must decide which details to simplify or abstract away.

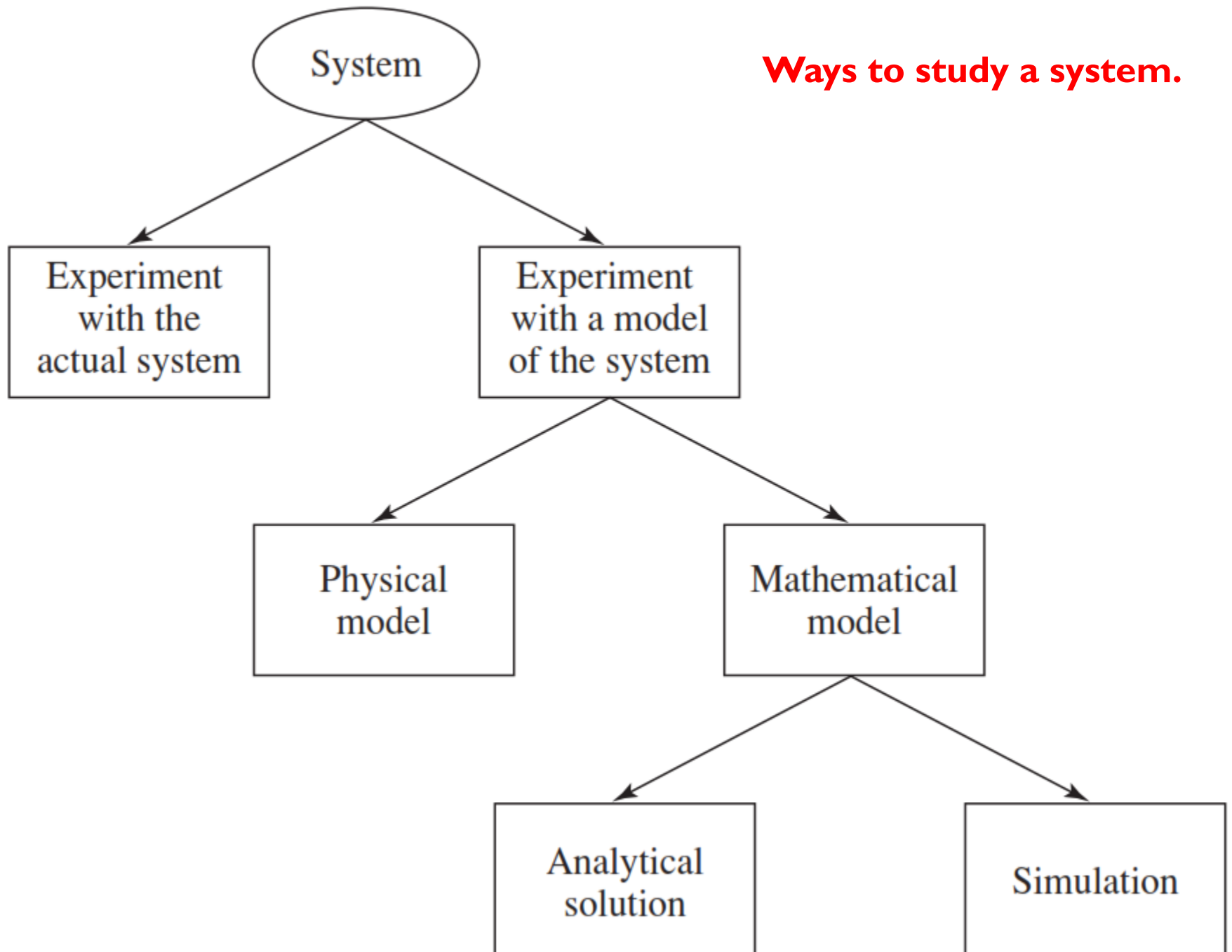


- **Model:** The result of abstraction - includes essential features. Can be diagrams/equations (for analysis) or computer programs (for simulation).
- **Simulation:** Imitating the operation of a real-world process or system over time.

GENERAL INTRODUCTION

- **Ex. Bank System:**
 - **System:** A real bank with customers, tellers, and queues
 - **Model:** A diagram showing customer flow, with equations for arrival and service rates (ignoring building color, furniture)
 - **Simulation:** A computer program that shows how queues grow from 9 AM to 3 PM

Ways to study a system.



GENERAL INTRODUCTION

- A **model** is a simplification of a real system.
- **Modeling** is the process of representing a system with a specific tool to study its behavior.
- **A model can be:**
 - ✓ **Analytic:** When a mathematical approach is feasible (e.g., formulas, equations)
 - ✓ **Simulation:** Used for complex systems (e.g., computer programs)
 - ✓ **Experimental:** When the real system already exists (e.g., prototypes)

GENERAL INTRODUCTION

- **Results can be:**
 - **Prediction:** What the system will do
 - **Explanation:** Why it behaves as it does
 - **Design:** Intended to achieve a purpose
- **Validation:** Comparing real-world measurements with results from analysis and simulation

Modeling:

Obtain a set of equations
(mathematical model) that
describes the behavior of the
system

A model describes the mathematical
relationship between inputs and
outputs

Simulation:

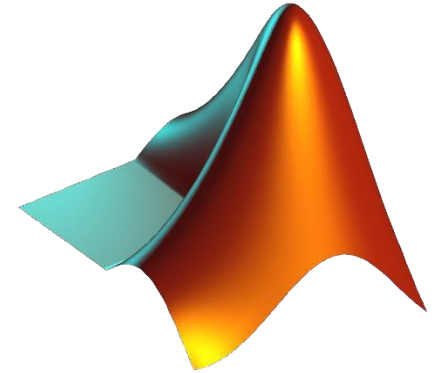
Use the mathematical
model to determine the
response of the system in
different situations.

GOAL OF MODELING AND SIMULATION

- A model can be used to investigate a wide variety of “**what if**” questions about real-world system.
 - ✓ Potential changes to the system can be simulated and predicate their impact on the system.
 - ✓ Find adequate parameters before implementation.
- **So, simulation can be used as**
 - ✓ Analysis tool for predicating the effect of changes.
 - ✓ Design tool to predicate the performance of new system.
- **It is better to do simulation before implementation.**

SIMULATION LANGUAGE

- **MATLAB®** - Excellent for mathematical modeling
- **Python** - Popular for general-purpose simulation with libraries like SimPy
- **Java** - Good for large-scale, object-oriented simulations
- **C/C++** - For high-performance simulations
- **Others ..** - Many other languages and specialized tools



SIMULATION SOFTWARE

■ Some of important:

✓ **AnyLogic**

✓ Arena

✓ AutoMod

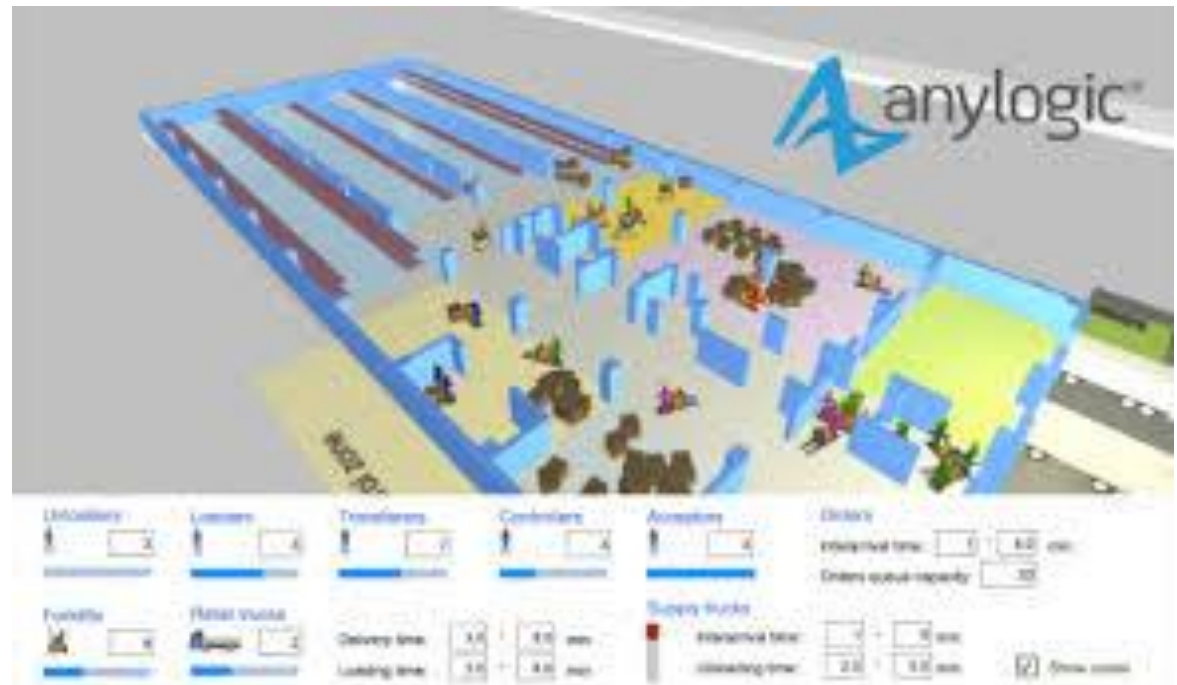
✓ ExtendSim

✓ Flexsim

✓ ProModel

✓ SIMUL8

✓ Others ...



WHEN SIMULATION IS THE APPROPRIATE TOOL?

- Study internal interactions of complex subsystems
- Simulate environmental changes and their effects
- Gain knowledge about system improvement
- Find important input parameters through variation
- Test new designs before implementation
- Determine machine requirements by testing different capabilities

WHEN SIMULATION IS THE APPROPRIATE TOOL?

- **Risky**: Risk involved in experimentation is another factor. In some systems there is a risk of damaging the system, or a risk of life.
- A plan can be visualized with animated simulation.
- The modern system is too complex that its internal interaction can be treated only by simulation.
- **Too expensive**: Experimenting with a real system is an extremely costly affair.
 - ✓ For example, the physical experimentation of a complex system like the satellite system is quite expensive and time- consuming.

WHEN SIMULATION IS NOT APPROPRIATE ?

- When the problem can be solved by common sense.
- When the problem can be solved analytically.
- If it is easier to perform direct experiments.
- If cost exceed savings.
- If resource or time are not available.

ADVANTAGES OF SIMULATION

- **Test hypotheses** about how/why phenomena occur
- **Compress or expand time** - speed up slow processes or slow down fast ones
- **Understand variable interactions**
- **Identify important variables** affecting performance
- **Test new designs easily** (hardware, layouts, transportation)
- **Understand system operation**
- **Answer "what if" questions** for new systems

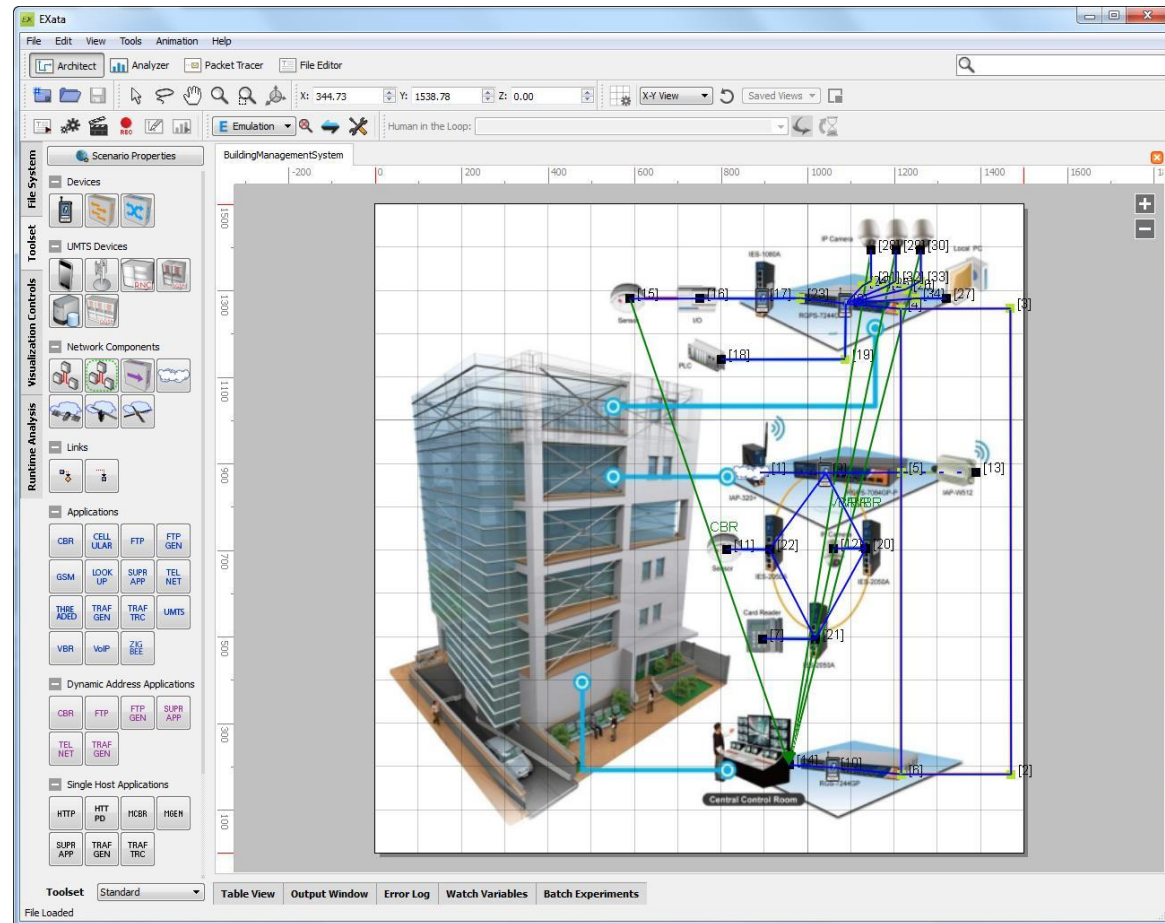
DISADVANTAGES OF SIMULATION

- Model building requires special training.
- Simulation results can be difficult to interpret.
- Simulation modeling and analysis can be time consuming and expensive.
- Simulation is used in some cases when an analytical solution is possible.

AREAS OF APPLICATION

■ I. Networks

Computer and communication networks



AREAS OF APPLICATION

■ 2. Military Applications

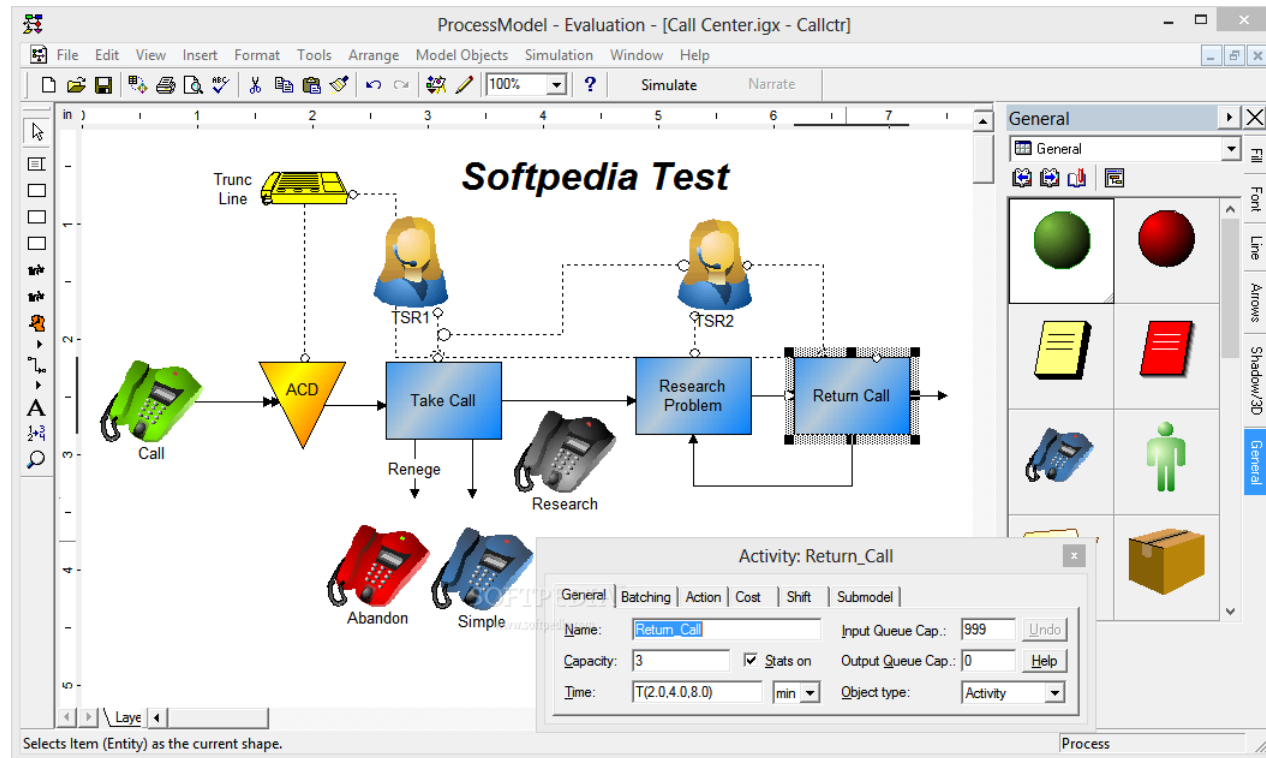
Training, strategy, equipment testing



AREAS OF APPLICATION

■ 3. Call Centers

Staffing levels, wait times



AREAS OF APPLICATION

■ 4. Manufacturing Applications

Production lines, inventory, scheduling



AREAS OF APPLICATION

■ 5. HealthCare

Patient flow, resource allocation



AREAS OF APPLICATION

■ 6. Road Traffic

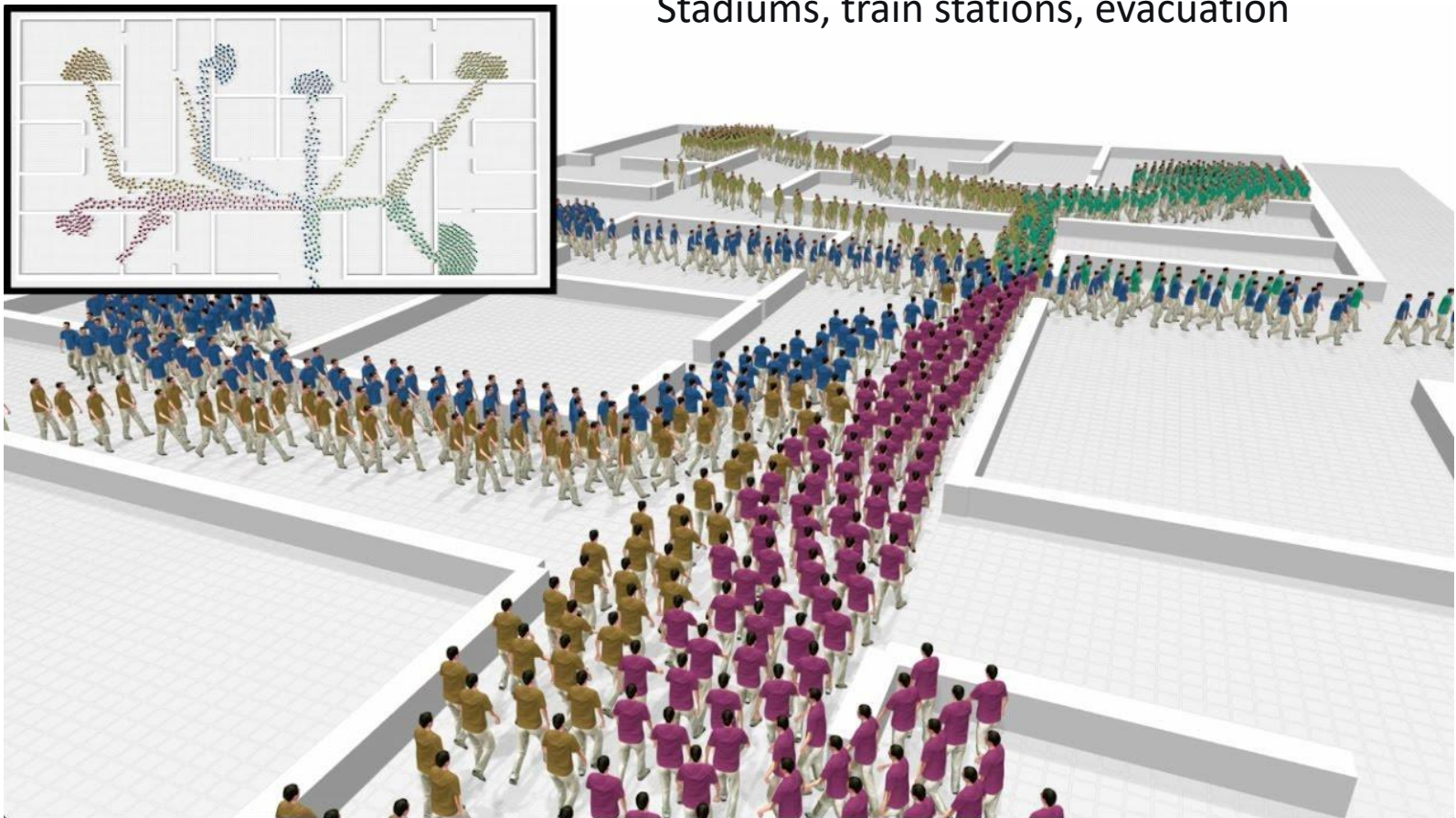
Congestion, traffic light timing



AREAS OF APPLICATION

■ 7. Crowd Flow

Stadiums, train stations, evacuation



AREAS OF APPLICATION

■ 8. Airport

Passenger flow, baggage handling



AREAS OF APPLICATION

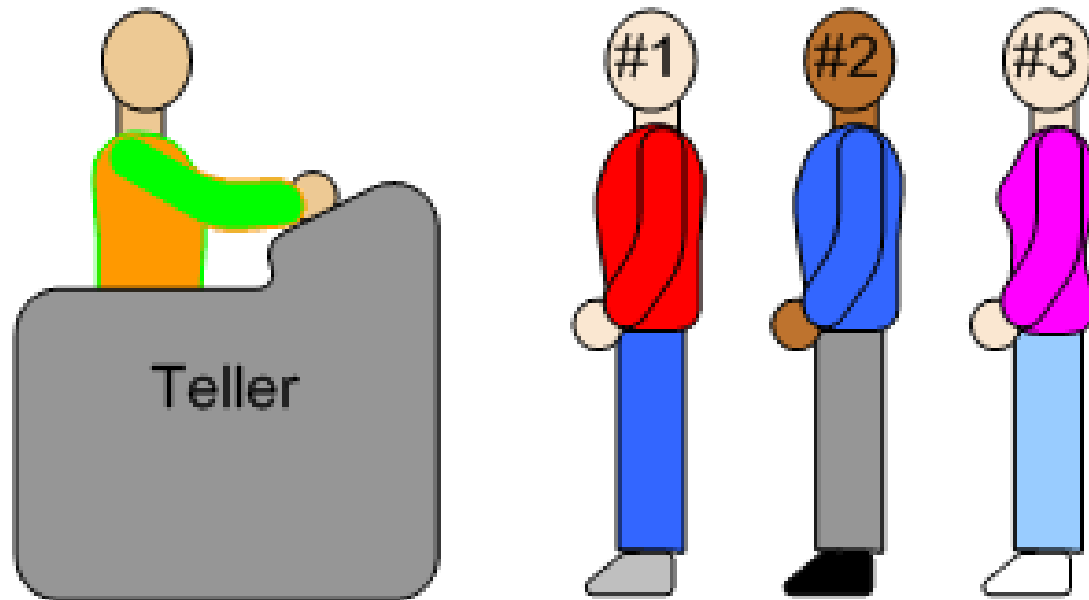
■ 9. Car Garage

Service scheduling, resource allocation



AREAS OF APPLICATION

■ 10. Banking



Teller staffing, queue management

AREAS OF APPLICATION

■ II.Trains Stations



AREAS OF APPLICATION

- **12. Highway Junction**



SYSTEMS

- **What a "system" means:**
- **From Greek "systema"** - organized relationship among functioning units
- Any set of interrelated components acting together for a common objective
- Group of objects joined in regular interaction toward a purpose
- A system is not just a collection of parts - the parts must interact and work toward a common goal.

•**System:** A car (engine, wheels, transmission work together for transportation)

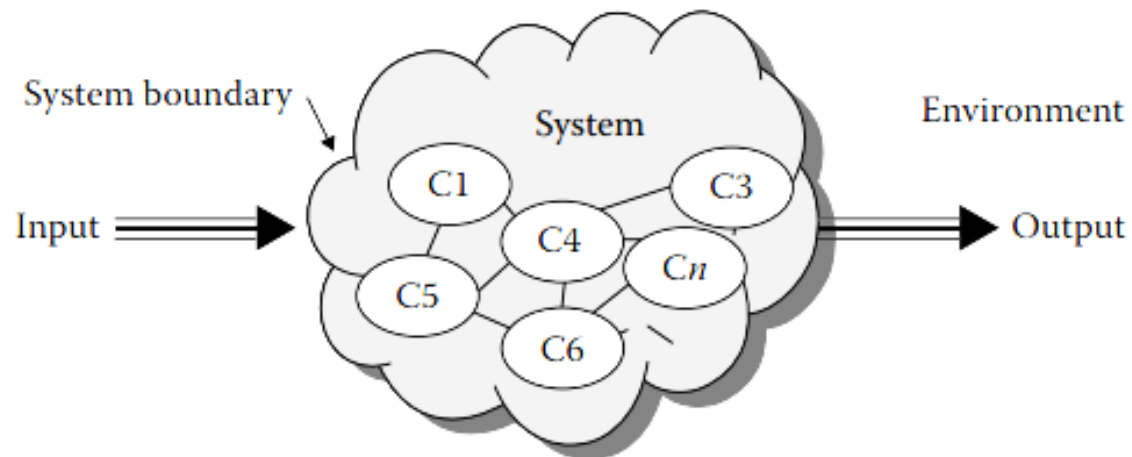
•**NOT a system:** Random people on the street (no common objective)

SYSTEMS

- The collection of entities that compose a system for one study might be **only a subset of another larger system.**
- System definition is not absolute - it's determined by what questions you're trying to answer.
 - **For example,**
 - **Study 1:** "How many tellers do we need?"
 - System = tellers + customers in teller area
 - **Study 2:** "How does loan processing affect overall bank performance?"
 - System = tellers + customers + loan officers + safety deposit

SYSTEM ENVIRONMENT

- **A system is characterized by the following attributes:**
 - ✓ **System boundary** - separates system from environment
 - ✓ **System components** - parts that make up the system
 - ✓ **Environment** - everything outside the boundary that affects the system

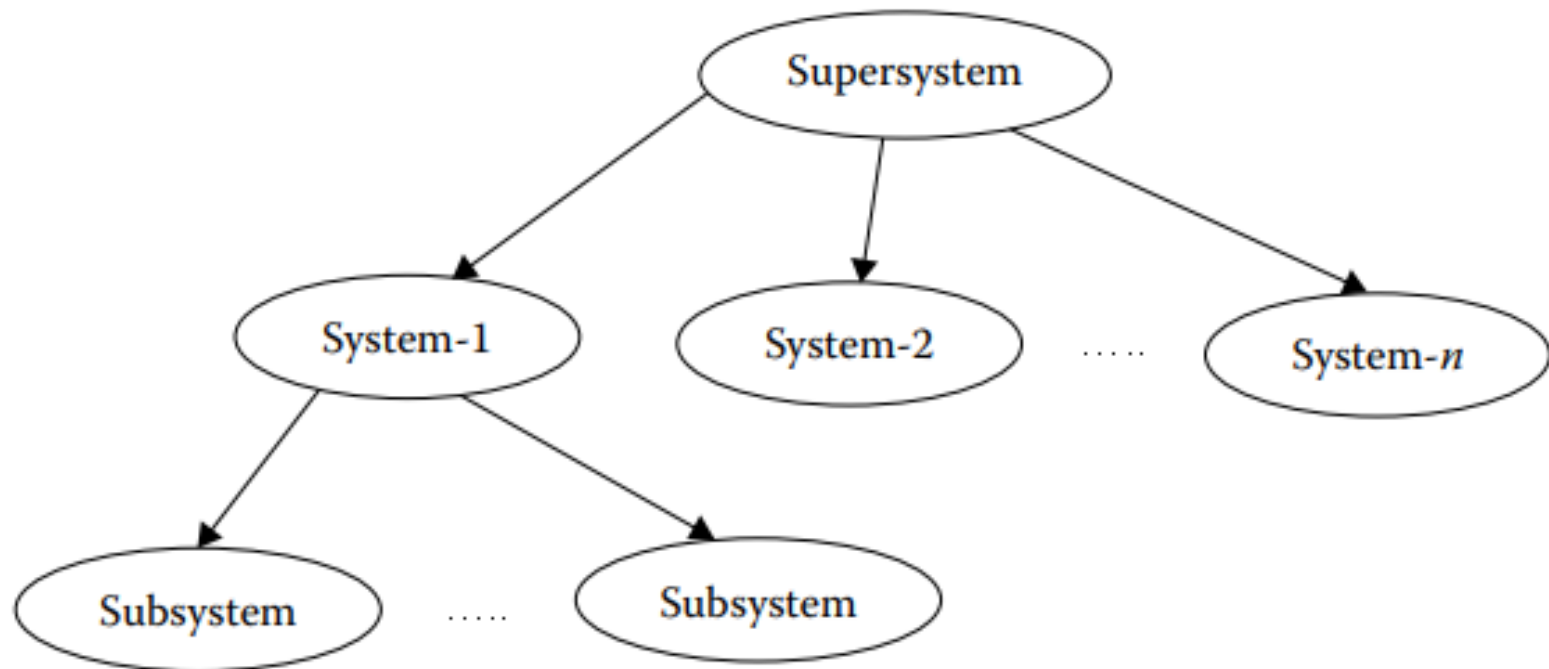


SYSTEM ENVIRONMENT

- A system is often affected by **changes occurring outside the system**. Such changes are said to occur in the system environment.
 - ✓ Factory :Arrival orders
 - ✓ Banks :Arrival of customers

SYSTEM ENVIRONMENT

- Every system consists of **subsystem** or components at lower levels and **supersystems** at higher levels.



COMPONENTS OF A SYSTEM

- An **entity** is an object of interest in the system.
 - ✓ Machines in factory.
 - ✓ Customers in Bank.
- An **attribute** is a property of an entity.
 - ✓ Speed, capacity.
 - ✓ Balance in customer accounts.
- An **activity** represents a time period of specified length.
 - ✓ Welding.
 - ✓ Making deposits.

Hospital:

- **Entities:** Patients, doctors, nurses, beds
- **Attributes:** Patient's age/condition, doctor's specialty, bed availability
- **Activities:** Examination (15 min), surgery (2 hours), recovery (3 days)

COMPONENTS OF A SYSTEM

- The **state** of a system is defined to be that collection of variables necessary to describe the system at any time, relative to the objectives of the study.
 - ✓ Status of machine (busy, idle, down,...)
 - ✓ In the study of a bank, possible state variables are the number of busy tellers, the number of customers waiting in line or being served, and the arrival time of the next customer.

COMPONENTS OF A SYSTEM

- **Event:** Immediate occurrence that might change system state
 - **Endogenous:** Occurs within system (service completion)
 - **Exogenous:** Occurs in environment (customer arrival)
 - ✓ In the bank study, the arrival of a customer is an exogenous event, and the completion of service of a customer is an endogenous event.

COMPONENTS OF A SYSTEM

- Examples of entities, attributes, activities, events, and state variables for several systems.

<i>System</i>	<i>Entities</i>	<i>Attributes</i>	<i>Activities</i>	<i>Events</i>	<i>State Variables</i>
Banking	Customers	Checking-account balance	Making deposits	Arrival; departure	Number of busy tellers; number of customers waiting
Production	Machines	Speed; capacity; breakdown rate	Welding; stamping	Breakdown	Status of machines (busy, idle, or down)
Communications	Messages	Length; destination	Transmitting	Arrival at destination	Number waiting to be transmitted
Inventory	Warehouse	Capacity	Withdrawing	Demand	Levels of inventory; backlogged demands



Thanks
Any questions